Project Documentation: Wireless Assistive Gaming Station for Natascha

- **Project Title:** Project Unwired: An Acceleration-Based, Dual-Node Wireless Assistive Controller for Wheelchair Integration
- **Short Description:** A wireless, modular assistive gaming station utilizing an acceleration-tracking joystick and a Bluetooth-based knee switch to enable independent PC gaming for a user with Cerebral Palsy.
- **Team Name:** Team Probello
- **Team Members :** Kerem Kaan Özmenteşe, Sandeep Sharma, Stiina Mäkelä
- **Flag / Logo:**

Figure 1 : Team Probello Flag

- **License:** MIT License
- **Estimated Cost of Solution:** €35.00

Figure 2 : Final Product mounted on Wheelchair

Contents

What Was the Problem?

The objective of this project was to refine and optimize an existing assistive gaming station prototype for our co-designer, Natascha. The target games are games like Blue Prince, an adventure, strategy, and puzzle-roguelike game played on PC. The game requires camera perspective control (mouse movement) and character navigation (WASD keys).

While a previous iteration successfully mapped these keybinds into an accessible hardware layout, their final design introduced physical challenges:

- The system was bulky and cumbersome.
- It relied on wiring spanning across the user's wheelchair setup.
- It used external AAA battery compartments and loose elastic bands, making it cumbersome to mount, position, and transport independently.

Natascha requested a streamlined, portable, and clean alternative that eliminated mechanical clutter and physical bulk.

Impairment & Functional Capabilities

To design an effective solution, the hardware structure had to align closely with the user's physiological realities:

- **The Impairment:** Natascha lives with Cerebral Palsy. This condition results in spasticity, involuntary tremors, and severe muscle stiffness. Consequently, conventional PC gaming mice, keyboards, and standard gamepads are inaccessible due to the fine motor precision and simultaneous button combinations they demand.
- **Residual/Retained Functions:** The system must leverage Natascha's existing physical strengths without causing rapid fatigue. Her core retained abilities include functional left-hand dexterity and controlled side-leg/knee movements.

Solution

Team Probello along with the co-designer brainstormed ideas and engineered a wireless, dual-node assistive setup that eliminates physical cable connections between the user's hand inputs, leg inputs, and the gaming PC. By treating hand inputs and leg inputs as isolated, self-powered wireless units, the controller drastically minimizes setup time and mitigates physical bulk on the wheelchair.

The architecture consists of two main components:

1. **The Acceleration-Detecting Joystick (Hand Node):** Housed in a custom shell tailored for Natascha's left-hand dexterity, this joystick contains an internal Microcontroller Unit (MCU) equipped with an accelerometer/IMU(Inertial Measurement Unit) sensor. It detects physical acceleration and emulates these movements directly into PC HID(Human interface Device) outputs (WASD keys and Mouse movement).

2. **The Puck.js Switch (Knee Node):** A standalone wireless Puck.js chip fitted into a custom mechanical fixture (developed by the previous HEIDI group working with Natascha - The Trentino Team). Positioned directly next to Natascha's wheelchair frame, it responds to her left knee/leg side movements.

Integrated Power & Ergonomic Upgrades

To ensure the hand node operates independently, the following electronic infrastructure was integrated directly into the 3D-printed handle:

- **Rechargeable Power:** Powered internally by a dedicated lithium polymer battery, discarding the previous group's loose external AAA battery packs.
- **Native Charging Loop:** A USB-C charging interface is exposed via a chassis cutout, allowing direct battery top-offs via a standard mobile phone cable.
- **Hardware Switch:** A physical toggle switch breaks the battery loop directly, enabling simple power-cycling and battery preservation between gaming sessions.

User Documentation

Operational Guide

Check : User Manual/README.md

Table of Controls and Associated Actions

The controller operates in two distinct modes, toggled dynamically by the user's leg movements to avoid overwhelming her with too many simultaneous inputs:

Physical Input Device	***User Action***	***Emulated System Output***	***Associated Game Action***
Puck.js Knee Switch	Single Press (Left Knee)	Mode Swap Toggle	Cycles between Keyboard Mode and Mouse Mode
Smart Joystick (Keyboard Mode)	Move Left / Right / Forward / Backward	Arrow Keys (W-A-S-D)	Controls character navigation
Smart Joystick (Mouse Mode)	Move Left / Right / Forward / Backward	Mouse Cursor Movement	Controls camera perspective and aiming

Technical Documentation

Bill of Materials (BOM)

- **1 x Microcontroller (MCU):** Seeed Studio XIAO nRF52840 (Features integrated Bluetooth LE and an internal LiPo battery charging/management circuit along with a native USB Type-C connector integrated directly onto the XIAO board).
- **1 x Knee Node Processor:** Puck.js Bluetooth Smart Beacon.
- **1 x Power Source:** HWE601525 3.7V / 170mAh rechargeable LiPo battery.
- **1 x Charging Extension:** USB-C Male-to-Female Extension Cable. Plugs into the internal XIAO port and extends out to the enclosure wall to create a rugged exterior charging socket
- **1 x Toggle Interface:** Single-Pole Single-Throw (SPST) sub-miniature slide switch.
- **1 x Interconnects:** Thin-gauge internal hookup wire fitted with a quick-disconnect wire connector plug at the power switch to allow clean mechanical teardowns.
- **Tools Required:** Soldering iron, wire strippers, heat-shrink insulation tubing, glue.

Reused Legacy Components (Group Trentino Assets):

- **1 x Joystick Cap:** Inherited and physically reused from the previous iteration's enclosure to secure the core mechanical alignment.
- **1 x Silicone Rubber Cover:** Reused and fitted over the top of the joystick interface to provide essential waterproofing and sweat resistance for daily user operation.
- **1 x Wheelchair Frame Mount Side Holder:** Physically retained from the previous setup to clamp the knee node securely onto the inner tubing of the wheelchair frame without needing to alter her existing wheelchair attachments.

Figure 3 : Puck.js (Used as Side Knee button)

Figure 4 : Housing for Side Knee Button (Reused part from Previous Team)

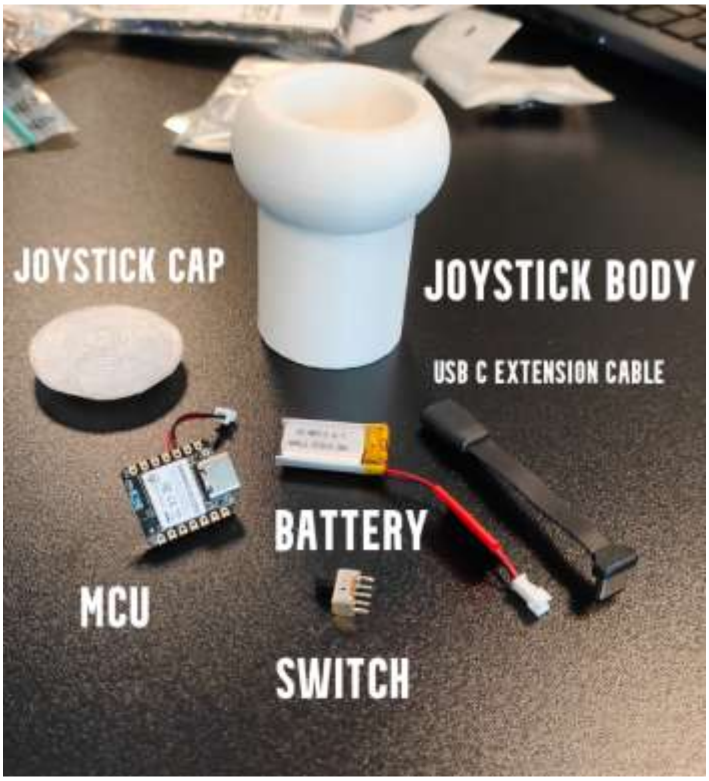

Figure 5 : Joystick body along with its components (without the silicone cap)

3D Printing Protocols & Slice Metrics

Check : CAD Files/READ ME - CAD Files and 3D Print.docx

Assembly Instructions

1. **Harness Prep:** Solder the negative lead of the 170mAh battery directly to the underside BAT- pad of the Seeed Studio XIAO.
2. **Switch Wiring:** Splice the quick-disconnect wire connector to the positive battery line. Route it directly to the terminals of the physical SPST slide switch, and solder the output lead to the BAT+ pad on the XIAO board.
3. **MCU Alignment:** Mount the wired XIAO board into its structural slots inside the 3D-printed hand enclosure, ensuring strict flush alignment between the USB-C charging interface and the outer casing boundary.
4. **Enclosure Closure:** Isolate all raw solder joints, secure the internal battery safely within its cradle, and snap/screw the upper housing cap down. Fit the Puck.js module into its respective wheelchair chassis clip.
5. **Joystick Cap:** Snapped tightly on top of the hand node assembly.
6. **Silicone Cover:** Placed over the joystick top for environmental protection.
7. **Wheelchair Frame Mount Side Holder:** Positioned firmly onto the inner frame of the wheelchair to provide the structural foundation for the knee switch.

Lessons Learned & Future Outlook

Future Improvements

- In the future, it would be beneficial to redesign the joystick's housing so that the hole placed over the wheelchair's control pin could be triangular or square instead of round. Natascha has already confirmed with the wheelchair manufacturer that the top of the control pin can be slightly sanded to help the joystick stay firmly in place.

Dongle Decision

- Originally, we also intended to use a dongle in our system, but we encountered difficulties programming it and integrating it with the other devices we were using. We therefore decided to leave it out of the project entirely. The dongle's original purpose was to reduce Bluetooth latency, but we found that the system performs fast enough even without it.

Future Multi-Device Expansion

During evaluation testing, Natascha noted that expanding this wireless architecture beyond games could drastically alter her day-to-day creative independence:

- **Stage Comedy Panel Integration:** Future updates can allow the wireless Puck.js knee switch to output MIDI(Musical Instrument Digital Interface) signals, letting Natascha cycle through and trigger physical audio soundboxes independently while performing live stand-up comedy on stage.
- **Universal Bluetooth Emulation:** Expanding the HID software protocols to let the acceleration-tracking joystick interface directly with commercial mobile applications, smart-home devices, and adaptive drone control applications.